



OPEN MedNet: a lightweight attention-augmented CNN for medical image classification

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Disease detection using medical images enables early and precise diagnosis. Despite the growing success of deep learning models, accurate classification remains a significant challenge. Medical images often exhibit characteristics such as limited spatial resolution, subtle visual differences between disease categories (low inter-class variance), and substantial variation within the same class (high intra-class variability). These factors collectively hinder the ability of standard vision models to generalize effectively, frequently resulting in misclassification. These challenges highlight the need for efficient architectures that can focus on critical spatial and contextual features for reliable performance. To mitigate these challenges we propose MedNet, a lightweight CNN architecture that combines depthwise separable convolutions with the CBAM attention mechanism to efficiently extract and refine spatially and contextually relevant features. The core ResidualDSCBAMBlock captures local patterns while CBAM enhances important spatial and channel-wise information, followed by adaptive pooling, dropout, and fully connected layers for robust classification. The model is trained and validated on the DermaMNIST, BloodMNIST, OCTMNIST which from MedMNIST, and Fitzpatrick17k datasets. MedNet matches or exceeds CNN baselines across these medical image datasets achieving higher accuracy with significantly fewer parameters and lower computational cost, demonstrating its effectiveness and efficiency in medical image classification tasks. Code is available at <https://github.com/Md-Ferdous/MedNet>.

Keywords MedNet, CBAM, MedMNIST, Fitzpatrick17k, ResidualDSCBAMBlock

Skin cancer, hematological conditions, and retinal pathologies are prevalent health concerns that demand timely and accurate diagnosis to prevent complications. Early detection plays a critical role in guiding effective treatment strategies. Medical imaging has emerged as a non-invasive and widely accessible modality for disease screening and diagnosis, offering critical support for early detection and treatment planning. The availability of DermaMNIST, BloodMNIST, and OCTMNIST from the MedMNIST collection¹, along with the Fitzpatrick17k dataset², has substantially accelerated research in automated diagnostic systems leveraging deep learning techniques. These datasets encompass a broad spectrum of medical imaging tasks, including skin lesion classification, blood cell morphology analysis, and retinal disease identification, providing standardized benchmarks that facilitate the comparison and evaluation of computer-aided diagnosis methods. Despite the advances enabled by these datasets, conventional Convolutional Neural Networks (CNNs) face significant challenges when applied to medical image analysis. Medical images frequently exhibit small spatial dimensions, subtle inter-class variations, and considerable intra-class variability, all of which complicate accurate classification and increase the risk of misdiagnoses. Furthermore, the deployment of deep learning models in clinical environments requires not only high predictive accuracy but also computational efficiency, particularly in resource-limited settings, such as portable devices or point-of-care diagnostic tools. These factors highlight the necessity for lightweight, interpretable, and computationally efficient architectures capable of focusing on the most informative and discriminative features within medical images. To address these challenges, attention mechanisms such as the Convolutional Block Attention Module (CBAM)³, SE-Net⁴, and BAM⁵ have been proposed to guide neural networks to emphasize salient spatial regions and channel-wise features, thereby improving classification performance, especially in the presence of noisy or cluttered backgrounds. In parallel, depthwise separable convolutions, introduced in MobileNet⁶ and Xception⁷, have demonstrated the ability to reduce the number of parameters and computational complexity while maintaining strong representational capacity. MedViT⁸ showed accuracy and AUC scores across all 12 MedMNIST classification datasets when

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compared with strong baseline models such as ResNet-18, ResNet-50, ConvNeXt, and other transformer-based methods. It offered a balanced trade-off between accuracy, robustness, and computational efficiency. The use of its hybrid architecture and the Patch Momentum Changer augmentation helped the model generalize better and remain more robust against variations and adversarial noise, demonstrating consistent performance across diverse medical image classification tasks. Biomedical images were resized to 224×224 and divided into 16×16 patches for input to the ViT-base-patch16-224 model⁹. Each dataset was trained separately using standard augmentation, normalization, and fine-tuning. Model performance was measured using accuracy and compared with existing benchmarks. The ViT achieved 97.9% on BloodMNIST, 90.4% on BreastMNIST, 94.6% on PathMNIST, and 57% on RetinaMNIST, outperforming previous methods. The results show that ViT effectively captures long-range spatial relationships and fine structural details in biomedical images. The combination of attention mechanisms with efficient convolutional operations offers a promising strategy for designing models that balance accuracy, efficiency, and interpretability. However, existing architectures rarely optimize these objectives jointly in the context of medical imaging datasets, leaving a gap for specialized models tailored to healthcare applications. In this work, we introduce MedNet, a compact yet powerful CNN architecture designed to address these challenges. MedNet integrates depthwise separable convolutions with CBAM within a residual learning framework, forming the core ResidualDSCBAMBlock, which efficiently captures local spatial patterns while enhancing discriminative feature representation. The network further incorporates adaptive pooling, dropout, and fully connected layers to ensure robustness against overfitting and variability in input data. Comprehensive experiments were conducted on four publicly available medical imaging datasets—DermaMNIST, BloodMNIST, OCTMNIST, and Fitzpatrick17k—covering a diverse set of classification tasks, from skin lesion detection to blood cell recognition and retinal disease diagnosis. Results indicate that MedNet consistently achieves competitive performance compared with state-of-the-art models, while substantially reducing both parameter count and computational overhead, demonstrating its potential as a solution for clinical applications. The key contributions of this work are summarized as follows:

- We propose MedNet, a lightweight and efficient CNN architecture that integrates depthwise separable convolutions and CBAM to improve classification accuracy on medical image datasets.
- We design a novel ResidualDSCBAMBlock that effectively captures spatial and contextual features while maintaining a low parameter count.
- We validate MedNet on multiple challenging datasets, where it outperforms existing models in terms of accuracy and efficiency, making it well suited for medical applications.

Background

The use of deep learning methods for medical image classification has garnered significant attention over the past decade, driven by the growing availability of medical image datasets and advancements in computational resources. Early approaches in automated skin lesion diagnosis utilized traditional image processing and handcrafted feature extraction methods, which often lacked robustness and scalability. With the advent of deep learning, CNNs have emerged as the de facto standard for medical image classification due to their ability to learn hierarchical feature representations. Models such as AlexNet¹⁰, VGGNet¹¹, and ResNet¹² have been adapted for dermatoscopic image analysis, achieving significant performance improvements. EfficientNet has emerged as an effective architecture for medical imaging tasks, which is proposed by Tan and Le¹³. EfficientNet introduces compound scaling to uniformly adjust network dimensions, thereby optimizing accuracy and efficiency. Skin disease classification, as in the DermaMNIST and Fitzpatrick17k datasets, has been widely explored using CNNs like ResNet¹², DenseNet¹⁴, and MobileNet⁶. Despite their effectiveness, these models often involve high computational costs and may struggle to generalize across diverse skin tones and lesion types, particularly in datasets like Fitzpatrick17k, which emphasizes skin tone diversity². Blood image classification using BloodMNIST has seen the application of lightweight models like EfficientNet¹³ and MobileNetV2¹⁵, which offer good performance with fewer parameters. While these models are computationally efficient, they may lack the spatial attention capabilities needed to distinguish visually similar blood cell types, especially in low-resolution images. Retinal disease classification, such as that performed on OCTMNIST, often benefits from deeper models like ResNet50 and Inception-based architectures¹⁶. These models capture fine-grained retinal structures well but tend to be resource-intensive, limiting their deployment in real-time clinical settings or on edge devices. To overcome the limitations of standard CNNs, attention mechanisms have been implemented to enhance model focus on the most relevant regions. To address classification tasks across different biomedical modalities, several deep learning architectures have been benchmarked, including both CNNs and vision transformers (ViTs). The study in¹⁷ implements a fine-tuned Vision Transformer (ViT-Base-Patch16-224) for 2D biomedical image classification on four subsets of MedMNISTv2, achieving notable accuracy: BloodMNIST (97.90%), BreastMNIST (90.38%), PathMNIST (94.62%), and RetinaMNIST (57%). The ViT model, pretrained on ImageNet and fine-tuned with dataset-specific hyperparameters, demonstrated its capability to generalize across diverse biomedical image types. Complementing this,¹⁸ conducted a comprehensive comparison between classical CNN models (AlexNet, VGG-16, ResNet-18, ResNet-50, DenseNet-121, EfficientNet-B0/B4) and ViT-based models (ViT-B/16, CLIP, DINO, SAM) on MedMNIST+. Their results revealed that CNNs, particularly DenseNet-121 and VGG-16, consistently outperformed ViTs in end-to-end training, highlighting the efficacy of simpler architectures when sufficient data is available. However, ViTs pretrained with DINO performed better in feature extraction tasks (e.g., linear probing and k-NN), showing promise for transfer learning applications. Addressing model robustness under corruption,¹⁹ introduced MedMNIST-C, simulating realistic image corruptions across 12 datasets using noise, blur, compression, and color shifts. They benchmarked classic architectures on both clean and corrupted data. ViT-B/16 achieved the highest robustness with relative balanced error (rBE) of 59.9%, outperforming CNN counterparts like VGG-16 (rBE 114%) and ResNet-50 (rBE 148%).

In²⁰, the authors revisited foundation models for medical image classification using CNNs and ViTs. DenseNet-121 achieved high performance on PneumoniaMNIST (ACC 94.02%), while ViTs like DINOv2 ViT-B/14 showed top results for DermaMNIST (ACC 92.02%, AUC 98.92%). SAM-C and CLIP variants of ViT further improved classification accuracy for datasets like PathMNIST and OrganAMNIST. Quantum machine learning has also been explored using MedMNIST datasets. The study in²¹ leveraged reduced-dimension (8×8) images for inference on IBM's 127-qubit quantum hardware. Results showed competitive performance (e.g., 85.4% ACC on PneumoniaMNIST), suggesting quantum circuits' potential for low-resource biomedical classification tasks. Lastly, MedMNIST v2²² introduced a comprehensive benchmark using both 2D and 3D datasets. While 2D CNNs demonstrated solid performance (70–90% ACC), AutoML tools (e.g., AutoKeras, Google AutoML) matched or exceeded manually tuned CNNs, showing the value of automated model selection in biomedical imaging. To further enhance model performance, attention mechanisms have been integrated into CNNs to improve feature focus and interpretability. Traditional attention modules like SE⁴ and CBAM³ introduced learnable channel and spatial attention to emphasize informative features. Medical datasets are frequently affected by class imbalance, where some lesion types are significantly underrepresented. This imbalance can lead to biased predictions and poor generalization. To mitigate this, Focal Loss²³ has been widely adopted. Focal Loss modifies the standard cross-entropy loss by down-weighting easy examples and focusing more on hard-to-classify cases, thus improving performance on minority classes. Collectively, these works provide a robust foundation for evaluating CNNs and ViTs on medical imaging tasks, while also pushing the frontier with robustness testing and quantum learning paradigms. The paper²⁴ introduces ERSR, a semi-supervised framework for fetal head segmentation in ultrasound images. It combines Dual-scoring Adaptive Filtering, Ellipse-constrained Pseudo-label Refinement, and Symmetry-based Consistency Regularization within a teacher–student setup to refine pseudo-labels and enforce stable shape learning. Leveraging the head's elliptical symmetry, ERSR achieves 95.36% Dice on HC18 with 20% labeled data and adds no inference overhead. IPA-CP²⁵, a semi-supervised framework for tumor segmentation in CT scans that targets small, hard-to-detect tumors. It uses uncertainty-based adaptive augmentation and iterative pseudo-label refinement within a mean teacher setup, plus a bidirectional copy-paste strategy for efficient data use. Evaluations on in-house and MSD datasets show superior accuracy and robustness over existing methods. PG-FANet²⁶ introduces a semi-supervised framework for histopathology image segmentation that integrates inter- and intra-uncertainty to enhance feature learning. Using a two-stage network with pseudo-mask guided feature enhancement and multi-scale aggregation, it refines segmentation accuracy. Evaluated on MoNuSeg and CRAG, PG-FANet outperforms state-of-the-art methods with limited labeled data. UniRepLNet²⁷ is a universal large-kernel ConvNet that unifies image, audio, video, point cloud, and time-series processing. Using Dilated Re-parameterization Blocks with small and large kernels, it efficiently captures global and local features. Scalable and modality-agnostic, it achieves state-of-the-art performance across diverse benchmarks. OverLoCK²⁸ is a pure ConvNet vision backbone inspired by human vision, combining an overview-first, look-closely-next attention mechanism. It includes Base-Net, Overview-Net, and Focus-Net, the latter using a Context-Mixing Dynamic Convolution (ContMix) for adaptive refinement through dynamic top-down context. With its Deep-stage Decomposition Strategy, OverLoCK achieves 84.2% Top-1 accuracy on ImageNet-1K, outperforming larger models while being more efficient across detection and segmentation tasks. PIF-Net²⁹ is a deep learning model for knee joint segmentation in MRI scans, featuring a parallel interweave fusion architecture that

Results

Dataset

Two publicly available datasets, MedMNIST v2 (2D)¹ and Fitzpatrick17k², were employed in the experiments. To examine the influence of input resolution, image sizes of 224×224 and 384×384 were selected. Larger input sizes provide increased spatial detail, allowing the attention layers to more effectively capture both the location (“where”) and the semantic content (“what”) of the salient regions. All images available in each dataset were utilized, without altering the dataset size and image also.

Implementation details

The model is trained for 70 epochs using the AdamW optimizer with a learning rate of 3×10^{-4} and a weight decay of 1×10^{-4} . A Cosine Annealing learning rate scheduler is employed with parameters $T_{\max} = 20$ and $\eta_{\min} = 1 \times 10^{-6}$ to adjust the learning rate during training. Focal Loss is used to handle situations where the data is imbalanced and some classes are harder to classify than others. Unlike standard Cross Entropy Loss, which gives the same importance to every training example, Focal Loss adds a factor $(1 - p_t)^\gamma$ that lowers the loss for easy, correctly predicted samples and increases it for hard, misclassified ones. This helps the model pay more attention to the less frequent or more challenging classes, instead of being influenced mainly by the majority class. The parameter γ decides how strongly the model focuses on difficult examples, while the optional α factor can give extra weight to minority classes. Because of this, Focal Loss is widely used in problems like medical image classification, anomaly detection, and object detection, where class imbalance is common and improving recall and sensitivity is essential. Training is performed with a batch size of 64. To ensure reproducibility, a random seed of 42, 43, and 44 is set for first, second, and third run, respectively. All the experiments are conducted on two RTX A6000 GPU with 48GB VRAM. The results are consistent across the validation and test set. Figures 1, 2, 3, and 4 illustrate the training and validation performance of different model configurations in 50 epochs, comparing CBAM-integrated and non-CBAM variants of the MedNet architecture on DermaMNIST, OCTMNIST, BloodMNIST and Fitzpatrick17k, respectively, at three input resolutions (384×384 , 224×224 , and 28×28). The top row shows the smoothed accuracy curves, while the bottom row presents the corresponding smoothed loss curves, with a smoothing factor of $\alpha = 0.3$ applied to reduce short-term fluctuations. In each plot, three independent training runs per configuration are displayed to assess run-to-run

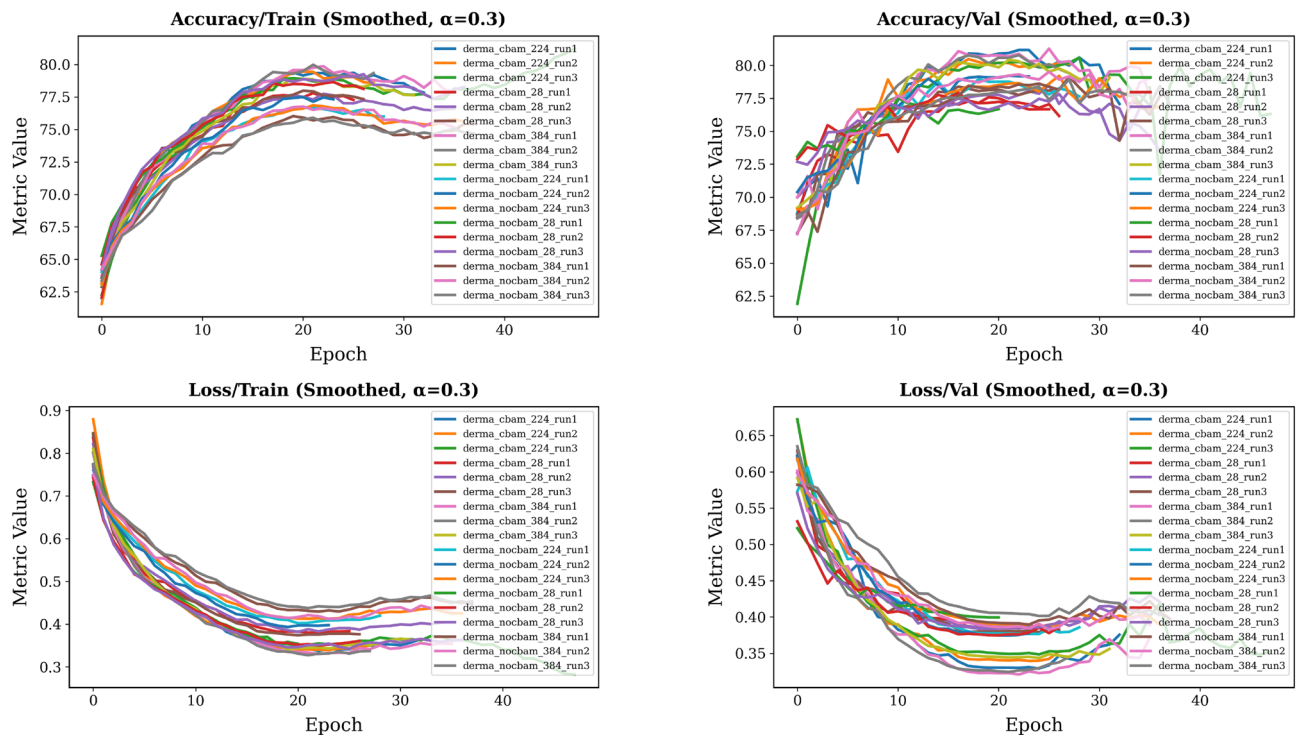


Fig. 1. Training and validation performance curves for the DermMNIST dataset. The plots illustrate (from left to right, top to bottom) the training accuracy, validation accuracy, training loss, and validation loss across epochs. Each curve represents one of three repeated experimental runs (run1, run2, and run3) performed under identical conditions to assess model stability and reproducibility. The smoothed curves (smoothing factor $\alpha = 0.3$) demonstrate convergence behavior and generalization performance.

consistency. In the training and validation accuracy plots (top left and top right), all models exhibit a steady increase in accuracy during the initial epochs. CBAM-based models generally achieve slightly higher accuracy across resolutions compared to their non-CBAM counterparts. The training and validation loss plots (bottom left and bottom right) show a consistent downward trend, confirming stable optimization behavior. Loss values decrease rapidly during early epochs and converge smoothly thereafter. The CBAM models demonstrate slightly lower final loss values, suggesting improved feature learning and generalization.

Table 1 presents a quantitative comparison of our models with the several existing deep learning and automated machine learning methods on three benchmark medical image datasets: DermMNIST, OCTMNIST, and BloodMNIST. Each result represents the mean and standard deviation across three independent runs, thereby ensuring the statistical reliability of the reported values. The table also includes architectural and computational characteristics of each model, such as the input image resolution, total number of parameters (in millions), and the computational cost measured in Giga multiply-accumulate operations (GMac). Where such data were unavailable in the original literature, “N/A” is used to indicate missing information. The first part of the table summarizes previously established baselines, including standard convolutional architectures such as ResNet-18 and ResNet-50, AutoML frameworks. For the ResNet models, two input resolutions- 28×28 and 224×224 -were considered. Increasing the input size generally leads to improved performance across all datasets but at the expense of higher computational cost. For example, in the case of ResNet-18, the FLOPs increase from 0.015 to 0.9 GMac when moving from 28×28 to 224×224 input resolution, accompanied by an improvement in accuracy on DermMNIST from 0.735 to 0.754. A similar trend is evident in ResNet-50, where higher-resolution inputs produce better results but demand substantially more computation (2.05 GMac). The automated learning frameworks, namely auto-sklearn, AutoKeras, and Google AutoML Vision, deliver competitive results without manual network design. Among these, Google AutoML Vision achieves relatively strong performance, with accuracies of 0.771 on OCTMNIST and 0.966 on BloodMNIST. However, to the best of our knowledge, these systems do not disclose architectural details or computational statistics, making direct comparisons with deep learning models difficult. Transformer-based methods are also included for comparison, such as the Vision Transformer (ViT) reported by⁹ and MedViT introduced by⁸. MedViT, which combines convolutional and transformer-based modules, achieves notable accuracy improvements, especially on DermMNIST (AUC = 0.937, ACC = 0.780), highlighting the potential of attention-based architectures in capturing global dependencies in medical imaging tasks. The lower half of the table reports the performance of the proposed model, evaluated both with and without the CBAM) across three input resolutions: 28×28 , 224×224 , and 384×384 . The results show that the model with CBAM consistently better performs both its non-attentive counterpart and all existing baselines. At 384×384 resolution, the model achieves the best overall results across all datasets, recording an AUC of 0.965 ± 0.0019 and ACC of 0.840 ± 0.0087 on DermMNIST, an AUC of 0.997 ± 0.0052 and ACC of

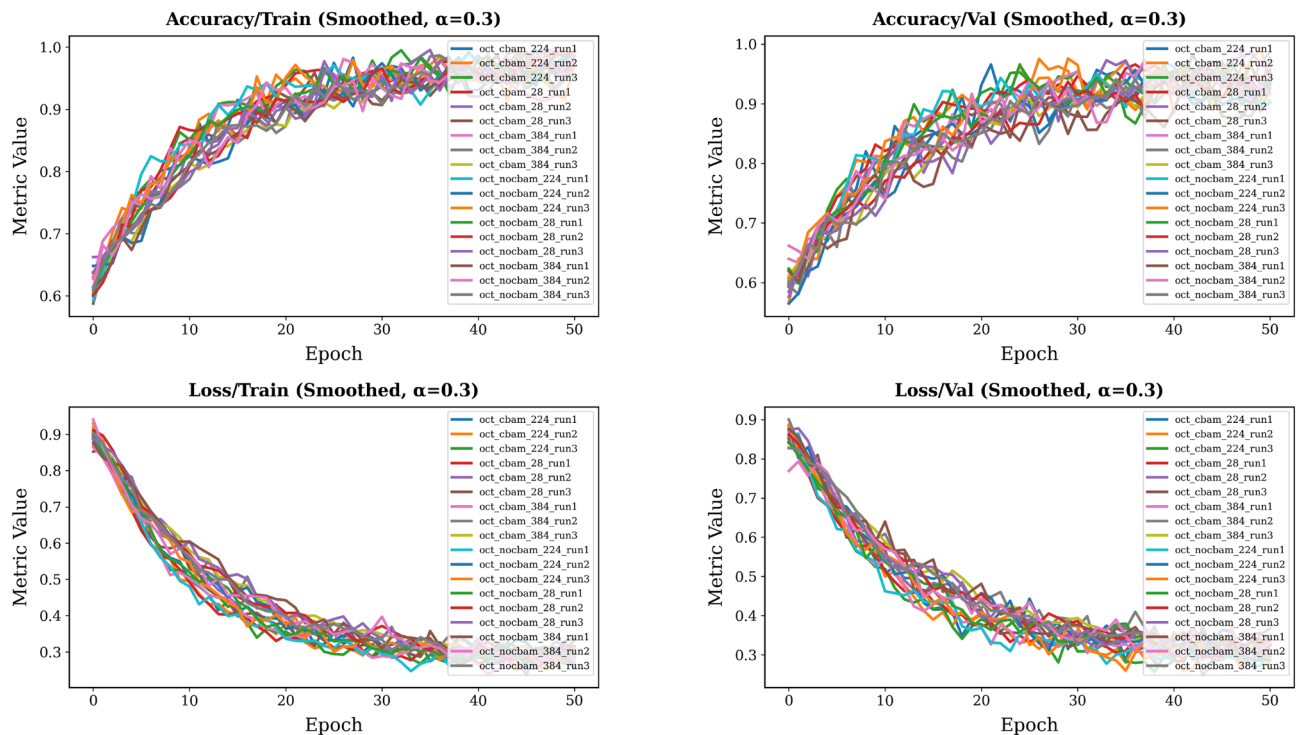


Fig. 2. Performance curves for training and validation on the OCTMNIST dataset. The plots show the training accuracy, validation accuracy, training loss, and validation loss over epochs (from top to bottom, left to right). To evaluate the stability and reproducibility of the model, three identically run experimental runs (run1, run2, and run3) are represented by each curve. Convergence behavior and generalization performance are demonstrated by the smoothed curves (smoothing factor $\alpha = 0.3$).

0.940 $\pm$ 0.0063 on OCTMNIST, and an AUC of 0.998 $\pm$ 0.0015 and ACC of 0.983 $\pm$ 0.0047 on BloodMNIST. These results indicate that both higher-resolution inputs and the inclusion of CBAM enhance the model's ability to extract fine-grained discriminative features, resulting in better classification performance. In contrast, the model without CBAM shows a decline in both AUC and accuracy, especially at higher input resolutions. For instance, on DermaMNIST at 384 $\times$ 384 input size, AUC decreases from 0.965 $\pm$ 0.0019 to 0.960 $\pm$ 0.0033 and accuracy from 0.840 $\pm$ 0.0087 to 0.800 $\pm$ 0.0044 when the CBAM module is removed. Similar degradations are observed across the other two datasets, suggesting that the CBAM mechanism plays a role in improving the model's representational capacity by focusing on informative spatial and channel-wise features. Furthermore, performance improvements become more pronounced as input resolution increases, implying that finer image details are better leveraged when combined with the attention mechanism. Overall, the results demonstrate that the proposed model achieves a strong balance between accuracy and computational efficiency. With only 3.27 million parameters and 5.9 GMac FLOPs at the highest resolution, it significantly outperforms much larger architectures such as ResNet-50 (25.6M parameters) and MedViT (23.6M parameters) while maintaining better classification performance across all three datasets. The findings also confirm the generalization ability of the proposed method, as it performs consistently well across diverse imaging modalities, including dermatology, optical coherence tomography, and blood cell microscopy. These observations emphasize that incorporating attention mechanisms like CBAM within a lightweight convolutional framework can lead to substantial gains in medical image classification accuracy without excessive computational demands.

Table 2 summarizes the training time and per-image inference time of our proposed models in a single 48GB VRAM, with and without CBAM attention module, on the DermaMNIST dataset at two different input resolutions. As expected, higher input resolution (384 $\times$ 384) results in a substantially longer training time compared to 224 $\times$ 224 inputs. For instance, the CBAM-enhanced model requires 10078.43 seconds at 384 $\times$ 384, whereas the same model trains in only 3968.8 seconds at 224 $\times$ 224. Similarly, the non-CBAM version shows shorter training times of 8059.98 seconds (384 $\times$ 384) and 3215.35 seconds (224 $\times$ 224). In terms of inference efficiency, all models achieve fast per-image processing. The inference time ranges from 0.00807 seconds (without CBAM, 224 $\times$ 224) to 0.01158 seconds (with CBAM, 384 $\times$ 384). The addition of CBAM introduces a slight increase in computational overhead, both during training and inference, but this increase remains modest given the performance improvements it provides. These results highlight that our models remain computationally efficient while achieving strong predictive performance, making them suitable for medical image analysis scenarios.

Table 3 presents the performance comparison on the Fitzpatrick17k dataset, where each experiment was conducted three times and the results are reported as the mean $\pm$ standard deviation. The table includes both three-class and nine-class classification tasks, evaluated in terms of AUC and ACC. The method proposed²

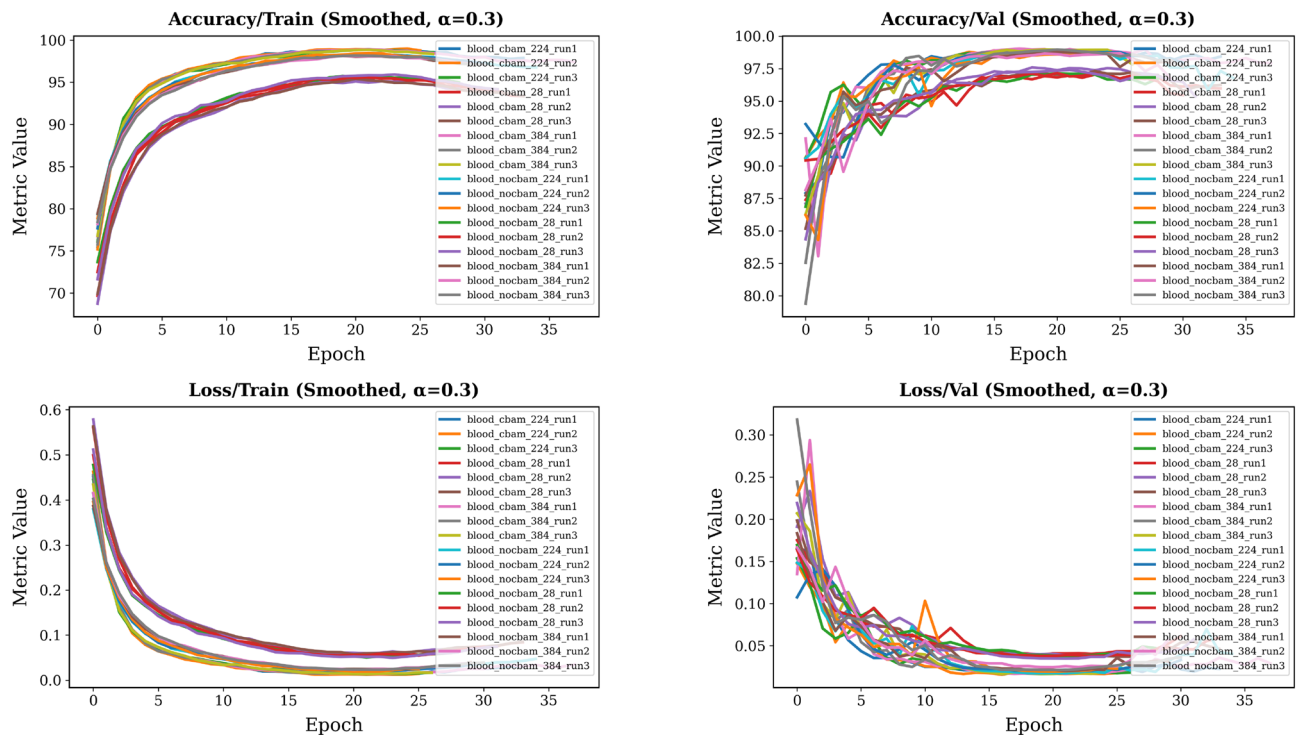


Fig. 3. Training and validation performance curves using the BloodMNIST dataset. Plots of training accuracy, validation accuracy, training loss, and validation loss over epochs (from left to right, top to bottom) are displayed. Each curve represents three identically run experimental runs (run1, run2, and run3) in order to assess the model's stability and reproducibility. The smoothed curves (smoothing factor $\alpha = 0.3$) show convergence behavior and generalization performance.

achieved an accuracy of 0.624 for the three-class task and 0.361 for the nine-class task, while the corresponding AUC values were not reported. In contrast, our proposed method was evaluated both with and without the CBAM at two different input resolutions: 224×224 and 384×384 . When CBAM was applied, the model achieved better performance compared to the version without CBAM. Specifically, for the three-class classification, the model with CBAM reached an AUC of 0.736 ± 0.0004 and 0.721 ± 0.110 at 224×224 and 384×384 input sizes, respectively. The corresponding accuracy values were 0.745 ± 0.0054 and 0.742 ± 0.0074 . For the nine-class task, the same model obtained AUC values of 0.729 ± 0.0041 and 0.724 ± 0.031 , with accuracies of 0.736 ± 0.0064 and 0.713 ± 0.0047 . The three-class classification accuracy improved to 0.745 ± 0.0054 and 0.742 ± 0.0074 , while the nine-class classification accuracy reached 0.736 ± 0.0064 and 0.713 ± 0.0047 for input sizes of 224×224 and 384×384 , respectively. The corresponding accuracies followed a similar trend. According to our study, the MedNet model performed well as reported in the Fitzpatrick17k dataset³³, and². In comparison, the model without CBAM showed slightly lower results across all metrics and input sizes.

Table 4 provides a detailed breakdown of our model's performance on the DermaMNIST dataset across individual classes, using precision, recall, and F1-score as evaluation metrics. The model showed excellent results on the mel and vasc classes, with F1-scores of 0.93 and 0.92, respectively, suggesting that it can reliably detect melanoma and vascular lesions. Performance on bcc, bkl, and nv was also solid, with F1-scores between 0.63 and 0.73. In contrast, the model struggled with the df class, where the F1-score dropped to 0.43. This drop is likely due to the small number of images in this category (only 23), which may have limited the model's ability to learn distinguishing features. Overall, the results show that the model handles the more common lesion types well, while performance on less represented classes remains a challenge due to class imbalance.

The classification reports in Tables 5, 6, 7 and 8 present the performance of our model across four medical imaging datasets: OCTMNIST, BloodMNIST, and the Fitzpatrick17k dataset (evaluated in both 3-class and 9-class formats). On OCTMNIST, the model performed exceptionally well across all four categories, achieving F1-scores between 0.90 and 0.98.

Similarly, in the BloodMNIST dataset, the model showed better performance across eight different blood cell types. Notably, it reached F1-scores of 1.00 for eosinophils and platelets, while even the lowest-performing class had an F1-score of 0.95. Performance on the Fitzpatrick17k dataset was more varied.

In the 3-class setting, the model achieved a strong F1-score of 0.85 for non-neoplastic lesions but struggled with the benign and malignant classes, which scored 0.20 and 0.43 respectively. These challenges became more apparent in the 9-class classification task, where certain categories—such as malignant dermal lesions—received F1-scores close to zero, largely due to the limited number of training examples. Still, the model performed well in the inflammatory category, which had the most data, achieving an F1-score of 0.81.

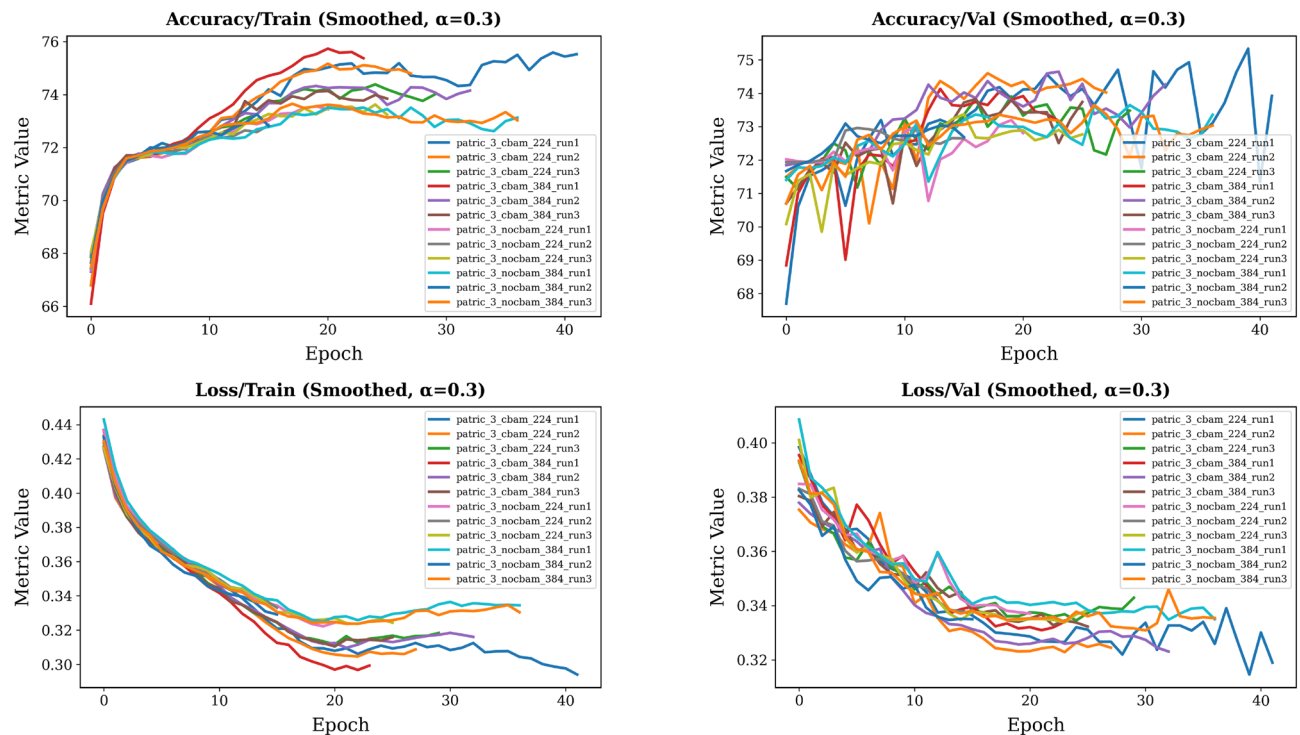


Fig. 4. Performance curves for training and validating the Fitzpatrick17k dataset. Across epochs, the plots show the training accuracy, validation accuracy, training loss, and validation loss from top to bottom and left to right. To evaluate the model's stability and reproducibility, three identically ran experiments (run1, run2, and run3) are represented by each curve. With a smoothing value of $\alpha = 0.3$, the smoothed curves exhibit both generalization and convergence behavior.

These findings highlight the model's strength when working with well-represented classes, while also emphasizing the need to address data imbalance in dermatological datasets.

Figure 5 shows a collection of correctly classified examples from the DermaMNIST dataset, covering all seven categories of skin lesions: akiec, bcc, bkl, df, mel, nv, and vasc. Each row in the figure represents a different class, and each image includes both the true and predicted labels, which align in every case. The images illustrate the wide range of visual features present within and between classes. For instance, bkl and nv lesions vary considerably in color and texture, while vasc lesions tend to have more distinct appearances. Despite these variations, the model was able to accurately identify each case, highlighting its ability to generalize across diverse lesion types. These examples underscore the importance of visual cues in skin lesion classification and demonstrate how automated systems can support diagnostic accuracy using dermoscopic images.

Figure 6 shows correctly classified illustrations from the OCTMNIST dataset. These include all four diagnosis types that are labelled; CNV, DME, DRUSEN and NORMAL. Every row is one class, and the color coding indicates if the true label equals the model prediction. The images from the OCT reveal some structural differences seen in retina conditions. In conditions like CNV and DME, the retinal layers appear thickened and distorted. DRUSEN cases have bumps or elevations underneath the retina, whereas a NORMAL scan is smooth and orderly without disruptions. These instances clearly exemplify how adeptly the model can outline the specific features unique to each condition. This bodes well for eye disease detection and diagnosis.

Figure 7 shows examples from the BloodMNIST dataset that the model classified correctly, covering all eight types of blood cells. There are three suitable examples in every class cell type represented in a row. The predicted labels are the same as the actual labels in each instance, which highlights the model's accuracy—even for similar looking cells. The model consistently identifies nuanced differences in cell size, shape, nuclear characteristics, and staining, as evident in monocytes, lymphocytes, and neutrophils. Being able to tell apart subtle differences in blood cell morphology, this model should help the automated analysis of blood smears. This can have useful applications in medical diagnoses.

Figure 8 presents a few skin lesion images that were correctly classified by the model. It gives the reader an insight into how it behaves with the data. The first row has normal moles and birthmarks and the model judged them correctly as benign. The images in the middle row show malignant lesions, they were less evenly shaped, sometimes crusty, or irregularly bordered. The model correctly recognized these as malignant, showing that it's able to pick up on the visual warning signs doctors look for when evaluating potential skin cancers. The bottom row shows non-cancerous but abnormal skin conditions such as psoriasis or other inflammatory rashes, which the model also got right. Overall, this set of examples highlights the model's ability to recognize key visual differences between harmless, dangerous, and inflammatory skin conditions.

Methods	Input size	Params (M)	FLOPs (GMac)	DermaMNIST		OCTMNIST		BloodMNIST	
				AUC	ACC	AUC	ACC	AUC	ACC
ResNet-18 (28) ³²	28×28	11.7	0.015	0.917	0.735	0.943	0.743	0.998	0.958
ResNet-18 (224) ³²	224×224	11.7	0.9	0.920	0.754	0.958	0.763	0.998	0.963
ResNet-50 (28) ³²	28×28	25.6	0.035	0.913	0.735	0.952	0.762	0.997	0.956
ResNet-50 (224) ³²	224×224	25.6	2.05	0.912	0.731	0.958	0.776	0.997	0.950
auto-sklearn ³²	N/A	N/A	N/A	0.902	0.719	0.887	0.601	0.984	0.878
AutoKeras ³²	N/A	N/A	N/A	0.915	0.749	0.955	0.763	0.998	0.961
Google AutoML Vision ³²	N/A	N/A	N/A	0.914	0.768	0.963	0.771	0.998	0.966
Vision Transformer ⁹	224×224	N/A	N/A	N/A	N/A	N/A	N/A	0.997	0.979
MedViT ⁸	224×224	23.6	4.9	0.937	0.780	0.960	0.782	0.997	0.951
Ours with CBAM	384×384	3.27	5.9	0.965 ± 0.0019	0.840 ± 0.0087	0.997 ± 0.0052	0.940 ± 0.0063	0.998 ± 0.0015	0.983 ± 0.0047
	224×224			0.953 ± 0.0074	0.81 ± 0.0066	0.982 ± 0.0063	0.926 ± 0.0058	0.985 ± 0.0051	0.974 ± 0.0018
	28×28			0.937 ± 0.0048	0.777 ± 0.0072	0.923 ± 0.0036	0.927 ± 0.0049	0.975 ± 0.0010	0.961 ± 0.0026
Ours without CBAM	384×384	3.09	5.89	0.960 ± 0.0033	0.800 ± 0.0044	0.964 ± 0.0051	0.921 ± 0.0041	0.994 ± 0.0072	0.978 ± 0.0068
	224×224			0.950 ± 0.0067	0.820 ± 0.0059	0.951 ± 0.0042	0.911 ± 0.0043	0.983 ± 0.0055	0.966 ± 0.0063
	28×28			0.890 ± 0.0043	0.758 ± 0.0075	0.963 ± 0.0046	0.924 ± 0.0068	0.976 ± 0.0013	0.952 ± 0.0018

Table 1. Comparison of different methods on DermaMNIST, OCTMNIST, and BloodMNIST datasets (3 repeated runs; mean±std).

Model	Image size	Total training time (s)	Per-image inference time (s)
Ours with CBAM	384×384	10078.43	0.01158
	224×224	3968.80	0.00877
Ours without CBAM	384×384	8059.98	0.01027
	224×224	3215.35	0.00807

Table 2. Model training and per-image inference time on the DermaMNIST dataset in a single 48GB VRAM. In this context, AUC refers to the Area Under the Curve, and ACC denotes Accuracy. N/A indicates that the corresponding value was not available or not reported in the reference.

Methods	Input size	3 classes		9 classes	
		AUC	ACC	AUC	ACC
Groh et al. ²	224×224	N/A	0.624	N/A	0.361
Ours with CBAM	224×224	0.736 ± 0.0004	0.745 ± 0.0054	0.729 ± 0.0041	0.736 ± 0.0064
	384×384	0.721 ± 0.0110	0.742 ± 0.0074	0.724 ± 0.031	0.713 ± 0.0047
Ours without CBAM	224×224	0.687 ± 0.0014	0.723 ± 0.0067	0.723 ± 0.0069	0.729 ± 0.0016
	384×384	0.689 ± 0.010	0.738 ± 0.0043	0.715 ± 0.0026	0.710 ± 0.0042

Table 3. Performance on the Fitzpatrick17k dataset (3 repeated runs; mean±std). N/A indicates that the corresponding value was not reported in the reference.

Class	Precision	Recall	F1-score	Support (test)
akiec	0.71	0.53	0.61	66
bcc	0.71	0.75	0.73	103
bkl	0.72	0.69	0.71	220
df	0.47	0.39	0.43	23
mel	0.93	0.93	0.93	29
nv	0.63	0.63	0.63	223
vasc	0.91	0.93	0.92	1341

Table 4. Classification report of our model on the DermaMNIST dataset.

Class	Precision	Recall	F1-score	Support (test)
CNV	0.83	1.00	0.90	250
DME	1.00	0.88	0.93	250
DRUSEN	1.00	0.87	0.93	250
NORMAL	0.96	1.00	0.98	250

Table 5. Classification report of our model on the OCTMNIST dataset.

Class	Precision	Recall	F1-score	Support (test)
Basophil	0.98	0.99	0.99	244
Eosinophil	1.00	1.00	1.00	624
Erythroblast	0.98	0.98	0.98	311
Immature granulocytes	0.95	0.96	0.95	579
Lymphocyte	0.96	0.99	0.97	243
Monocyte	1.00	0.91	0.95	284
Neutrophil	0.96	0.97	0.97	666
Platelet	1.00	1.00	1.00	470

Table 6. Classification report of our model on the BloodMNIST dataset.

Class	Precision	Recall	F1-score	Support (test)
Benign	0.41	0.14	0.20	300
Malignant	0.74	0.30	0.43	338
Non-neoplastic	0.76	0.94	0.85	1634

Table 7. Classification report of our model on the Fitzpatrick17k dataset for three classes.

Class	Precision	Recall	F1-score	Support (test)
Benign dermal	0.41	0.10	0.16	152
Benign epidermal	0.28	0.08	0.12	117
Benign melanocyte	0.32	0.13	0.18	31
Genodermatoses	0.47	0.09	0.15	142
Inflammatory	0.71	0.92	0.81	1492
Malignant cutaneous lymphoma	0.28	0.07	0.11	29
Malignant dermal	0.00	0.00	0.00	19
Malignant epidermal	0.62	0.42	0.50	204
Malignant melanoma	0.52	0.49	0.50	86

Table 8. Classification report of our model on the Fitzpatrick17k dataset for nine classes.

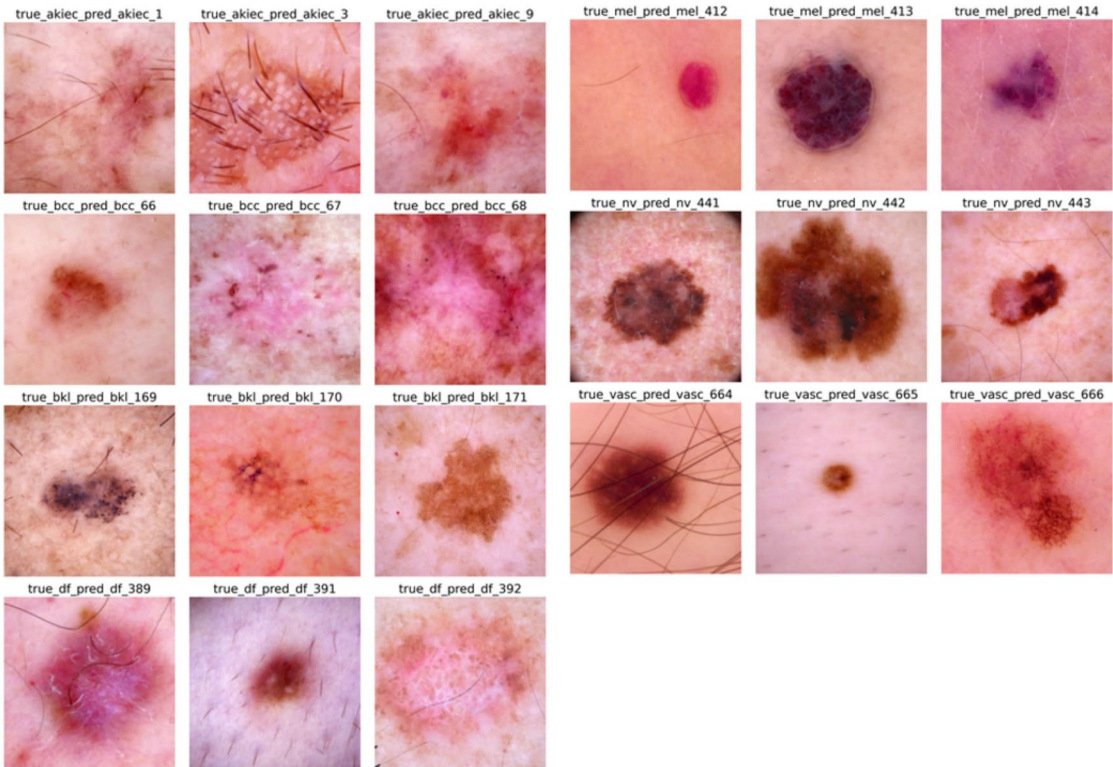


Fig. 5. Several true prediction of DermaMNIST.

Figures 9, 10, 11 and 12 represents Grad-CAM visualization across progressive processing stages are shown for DermaMNIST, OCTMNIST, BloodMNIST, and Fitzpatrick17k samples, respectively, illustrating how the model’s attention evolves through five sequential processing stages (Stage 1–Stage 5). Each column indicates the model’s processing stage, while each row presents class-specific activation patterns. Warmer color regions (red–yellow) indicate areas of higher model attention, and cooler colors (blue) correspond to regions of lower activation. The visualization highlights the model’s interpretability and its ability to develop class-specific focus patterns through successive learning stages. As the model progresses through the stages, its attention becomes

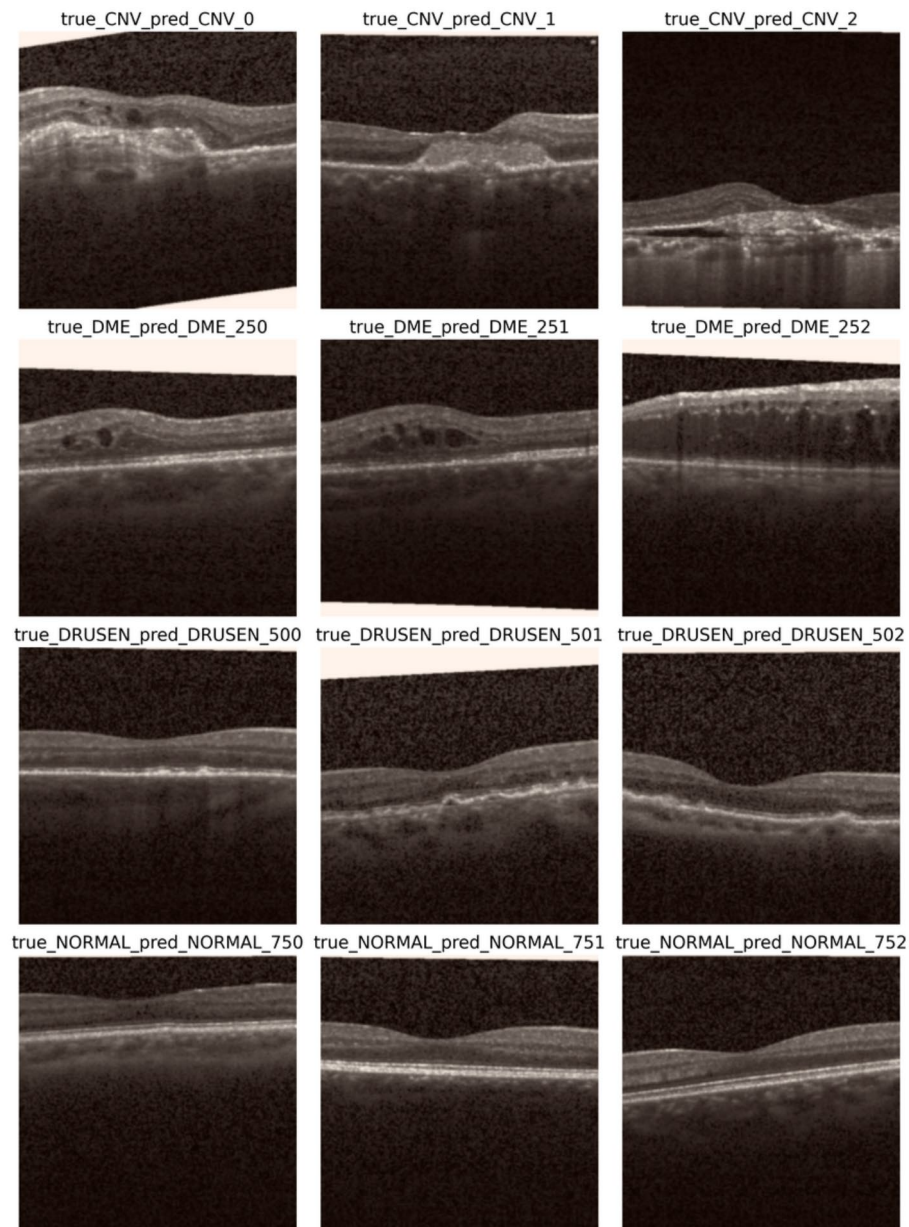


Fig. 6. Several true prediction of OCTMnist.

increasingly focused on clinically relevant regions by capturing finer texture and boundary details associated with disease progression.

Figure 13 shows several real examples where the model made incorrect predictions, helping to understand its limitations in practice. In skin images, it sometimes confused different types of lesions—for instance, mistaking a dangerous melanoma for a benign mole, or identifying a vascular lesion as something harmless. These types of errors are especially concerning in a clinical setting, where an incorrect diagnosis can delay critical treatment. The model also struggled with blood cell images, misclassifying cells that look quite similar under a microscope, like basophils and immature granulocytes. In eye scans, it failed to tell the difference between conditions like choroidal neovascularization and drusen, which can appear visually alike but require different treatments. These mistakes suggest the model relies too heavily on surface-level patterns and may miss deeper, medically important features—pointing to the need for better training data, more robust models, and closer collaboration with clinical experts.

Discussion

The results presented in Tables 1 and 3 underscore the effectiveness of our proposed model, MedNet, addressing the challenges of medical image classification—namely, small image sizes, low inter-class variance, and high intra-class variability. Despite having significantly fewer parameters (3.27M) and relatively low computational complexity (5.9 GMac), MedNet consistently well performed than several state-of-the-art models, including

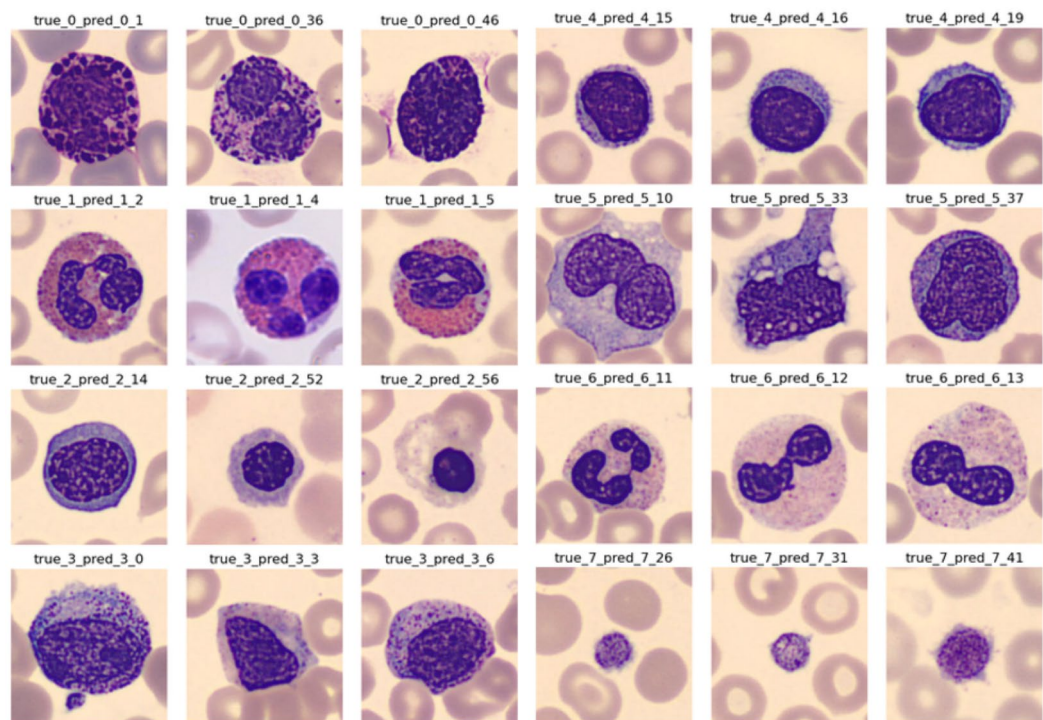


Fig. 7. Several true prediction of BloodMNIST.

ResNet-18, ResNet-50, and multiple AutoML-based solutions, across four benchmark datasets: DermaMNIST, BloodMNIST, OCTMNIST, and Fitzpatrick17k. On the three MedMNIST datasets, MedNet achieved the highest AUC and accuracy scores. Specifically, it reached an AUC of 0.965 ± 0.0019 and an accuracy of 0.840 ± 0.0087 on DermaMNIST, 0.997 ± 0.0052 AUC and 0.940 ± 0.0063 accuracy on OCTMNIST, and 0.998 ± 0.0015 AUC and 0.983 ± 0.0047 accuracy on BloodMNIST. These results are particularly notable when compared with deeper models like ResNet-50, which, while achieving competitive performance, do so at the cost of significantly higher computational requirements (e.g., up to 2.05 GMac). By integrating depthwise separable convolutions with the CBAM in the ResidualDSCBAMBlock, the network is able to capture both local features and contextual information. The CBAM mechanism emphasizes informative spatial and channel-wise regions, helping the model focus on relevant patterns despite noise or variability in the images. Furthermore, the evaluation on the Fitzpatrick17k dataset—designed to challenge models with real-world diversity in skin tones and lesion types—demonstrates the robustness of our approach. MedNet achieved an AUC of 0.736 ± 0.0004 and an accuracy of 0.745 ± 0.0054 in the 3-class task (benign, malignant, non-neoplastic), and an AUC of 0.729 ± 0.0041 and 0.736 ± 0.0064 accuracy in the more granular 9-class task. These results validate the model's generalization capabilities in dermatological datasets. Moreover, it is also important to highlight the trade-offs between model size, resolution, and performance. While increasing the input resolution from 28×28 to 224×224 slightly improved the performance of ResNet models, the computational cost increased also. In contrast, MedNet, trained at a higher resolution of 384×384 , maintained both high performance and computational efficiency—underscoring the use of its lightweight and attention-augmented design. Although we use Focal Loss to address the class imbalance problem, our model still shows lower performance on under-sampled image sets. In future work, we plan to explore strategies such as under-sampling, weighted sampling, or generative data augmentation to better handle this imbalance problem. Overall, the need for compact, interpretable, and high-performing CNN architectures tailored to the challenges of medical imaging. MedNet addresses this need effectively, offering a solution for limited computational resources.

Methods

The proposed architecture is designed for medical image classification task. It incorporates depth-wise separable convolutions and CBAM to improve feature extraction and attention-based learning. Additionally, a Focal Loss function is used to address class imbalance problem. The entire model is trained using optimized data augmentation techniques based on Albumentations to enhance generalization.

Data preprocessing and augmentation

To enhance the robustness and generalization ability of the model, a series of data augmentation and preprocessing operations were applied to the training images using the Albumentations library. To increase data diversity, each image has a 50% probability of being flipped horizontally and vertically. Moreover, to improve the model's robustness at different lighting conditions the brightness and contrast of the image alters with a 20% probability. In addition to this, we randomly shifts the image by up to 10% of its height or width, scales it within $\pm 10\%$,

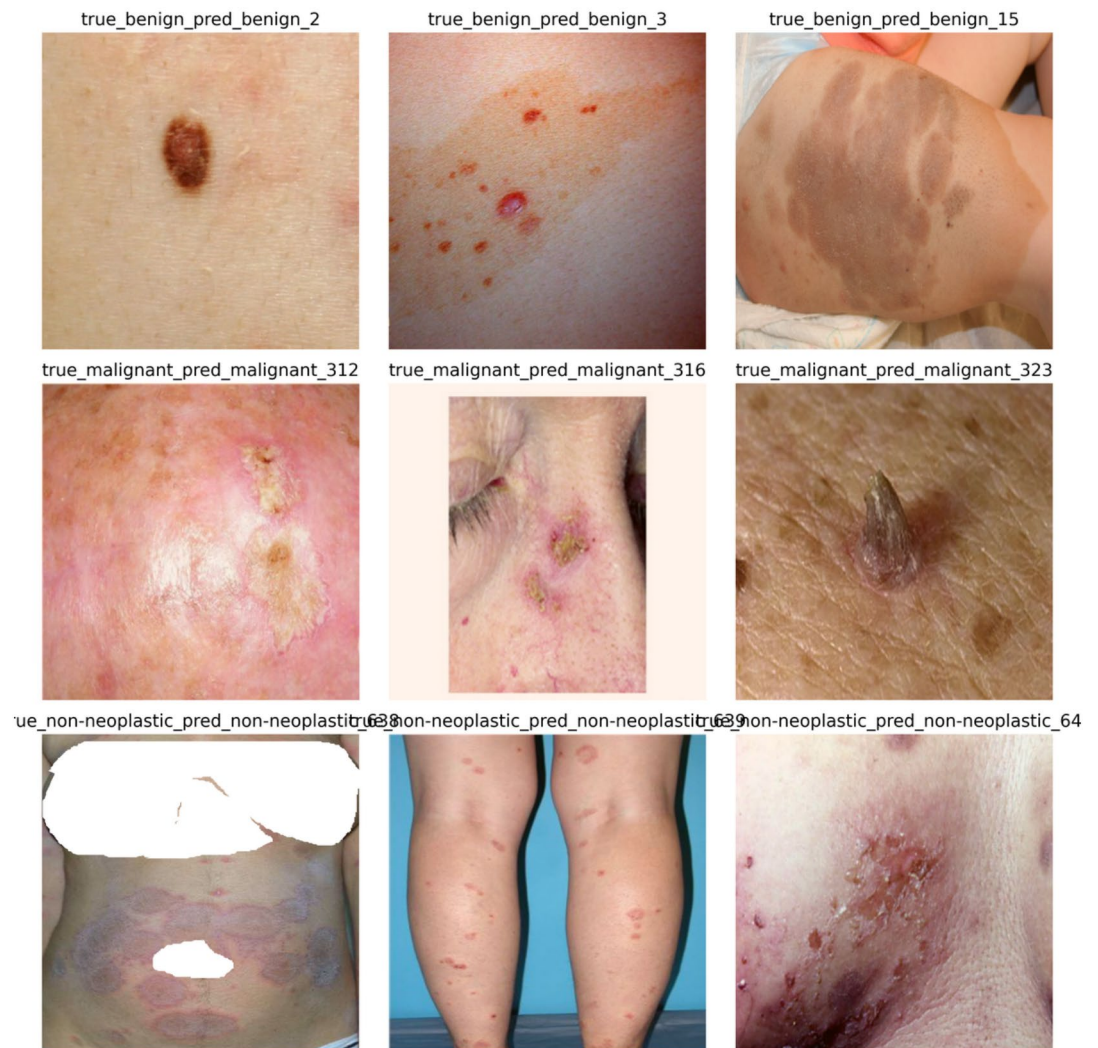


Fig. 8. Several true prediction of Fitzpatrick17k.

and rotates it by up to ± 20 degrees with a 50% chance. This helps the model to learn invariant features under spatial transformations. To improve visibility of fine details Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied with 20% probability.

Model architecture

Figure 14 shows the detailed breakdown of the proposed MedNet architecture and its components. We build our model using Depthwise Separable Convolutions layer and CBAM attention mechanism. ResidualDSCBAMBlock serves as the feature extractor of the network. This block is built with two consecutive depthwise separable convolution layer to extract efficient and spatial relevant features, followed by the CBAM attention module which refines these features by focusing on important channels and spatial regions via attention mechanisms. After this residual connection is added even if input/output shapes differ. Where, channel dimensions are matched using a 1×1 convolution and spatial dimensions are matched using stride ≥ 1 . The two Depthwise Separable Convolutions extract hierarchical spatial features and combine local information which are parameter efficient but may lose global context. Therefore, adding CBAM after this ensure context-aware refinement by learning which spatial and channel-wise areas matter most. Following the backbone, an adaptive average pooling layer is applied to the feature maps. To reduce overfitting, a dropout layer with a dropout probability of 0.4 which, randomly deactivate neurons and acts as a regularizer. After that, the pooled features are passed through two fully connected layers, first linear layer reduces the feature dimension from 1024 to 256, and the second layer maps the 256 dimensional vector to the number of output classes for final classification.

The proposed MedNet architecture is composed of five sequential stages, followed by pooling and fully connected layers, as detailed in Table 9. Each stage is constructed using a ResidualDSCBAMBlock, which integrates depthwise separable convolutions with residual connections and the CBAM to enhance feature representation. Stage 1 employs 64 filters with a stride of 1, and its ResidualDSCBAMBlock is expanded in the table to illustrate its internal configuration. The block consists of two convolutional layers (a 3×3 convolution followed by a 1×1 convolution), batch normalization, and a CBAM attention layer. ReLU activation is applied

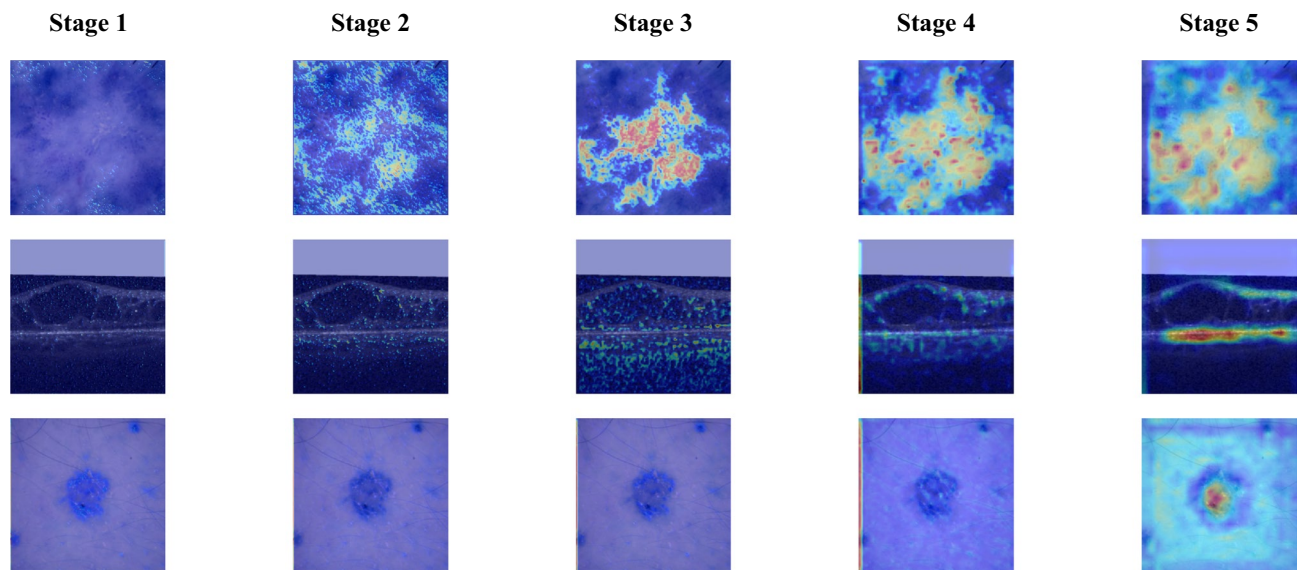


Fig. 9. Visualization of Grad-CAM heatmaps across five progressive stages (Stage 1–Stage 5) for three representative DermaMNIST samples corresponding to different disease classes. The first row denotes the processing stages, while the following rows illustrate class-specific activation patterns highlighting the model's focus at each stage.

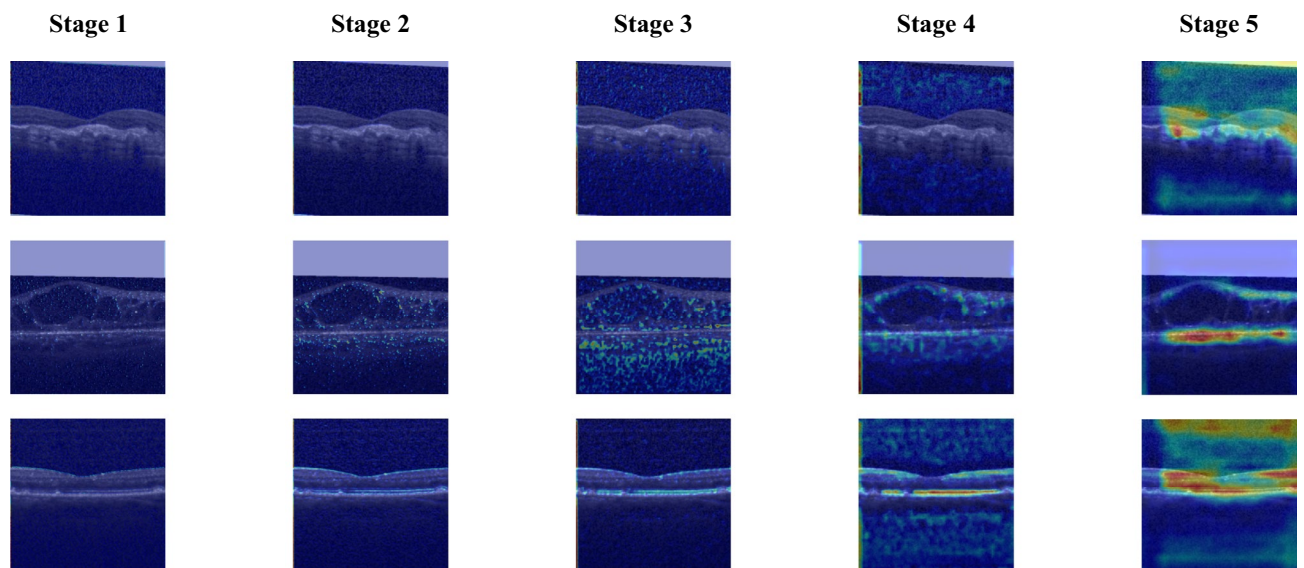


Fig. 10. Visualization of Grad-CAM heatmaps across five progressive stages (Stage 1–Stage 5) for three representative OCTMNIST samples corresponding to different disease classes. The first row denotes the processing stages, while the following rows illustrate class-specific activation patterns highlighting the model's focus at each stage.

after each convolution. Stages 2–5 progressively increase the number of filters (128, 256, 512, and 1024, respectively) with a stride of 2, enabling hierarchical feature extraction at multiple spatial resolutions. After Stage 5, an adaptive average pooling layer reduces the feature maps to a fixed size regardless of the input resolution. The pooled features are passed through two fully connected layers: Fc1 with 512 units and Fc2, which outputs predictions corresponding to the number of classes (n_c). This hierarchical design enables MedNet to capture fine-grained spatial features in earlier layers while progressively learning more abstract and semantically rich representations in deeper stages. The incorporation of CBAM attention further guides the network to focus on the most informative regions and channels, thereby enhancing classification performance.

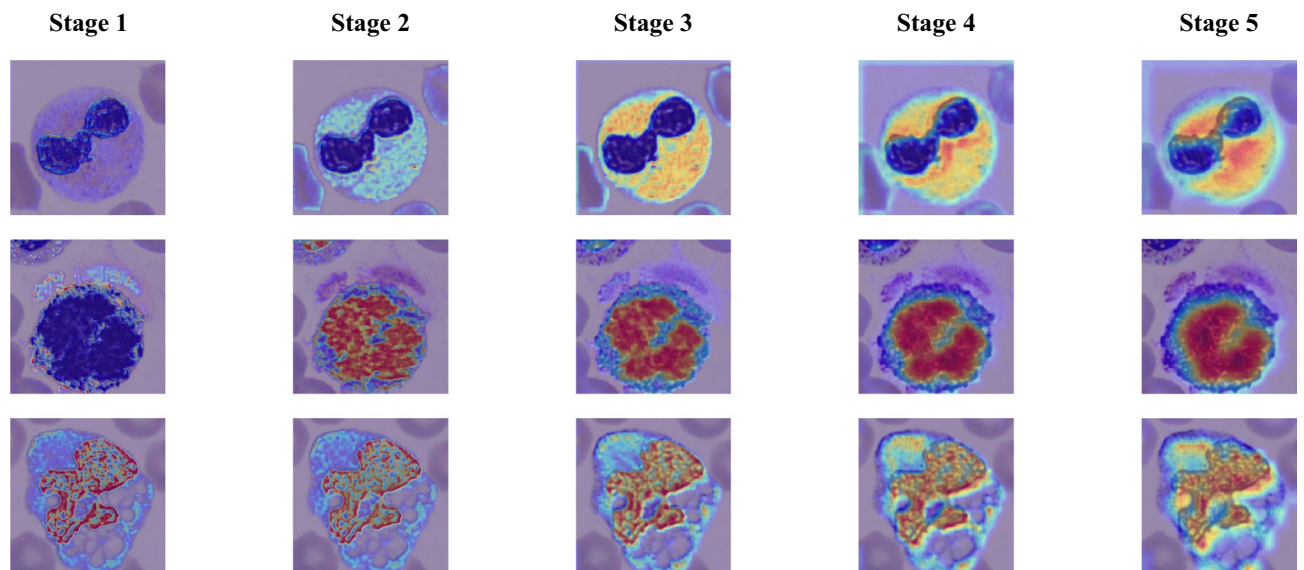


Fig. 11. Visualization of Grad-CAM heatmaps across five progressive stages (Stage 1–Stage 5) for three representative BloodMNIST samples corresponding to different disease classes. The first row denotes the processing stages, while the following rows illustrate class-specific activation patterns highlighting the model's focus at each stage.

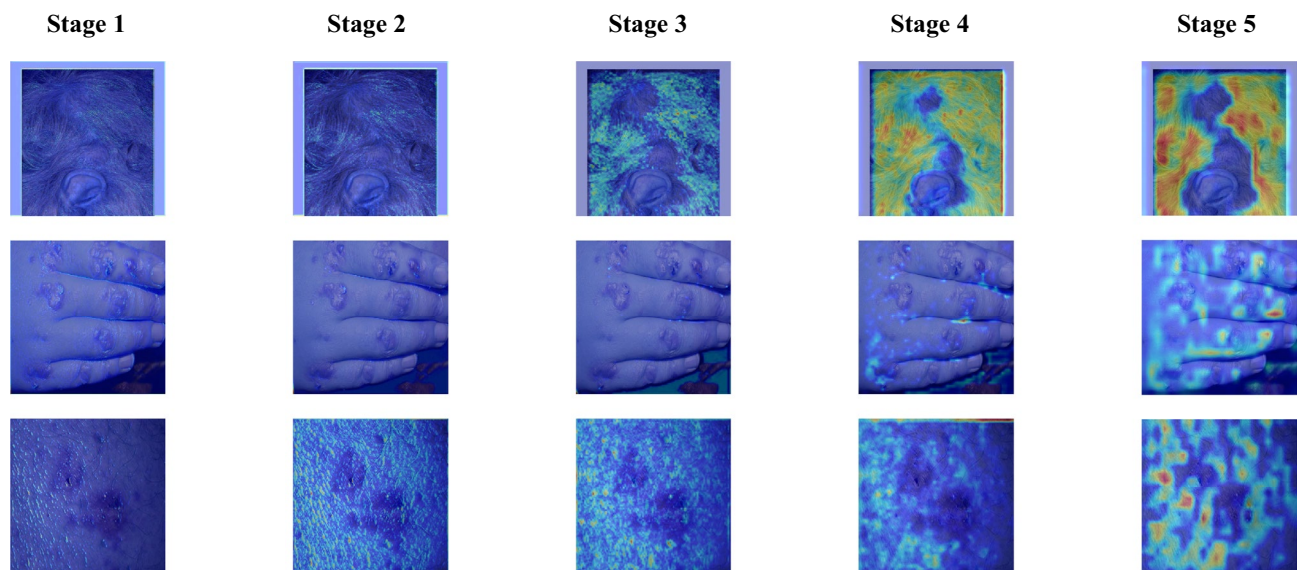


Fig. 12. Visualization of Grad-CAM heatmaps across five progressive stages (Stage 1–Stage 5) for three representative Fitzpatrick17k samples corresponding to different disease classes. The first row denotes the processing stages, while the following rows illustrate class-specific activation patterns highlighting the model's focus at each stage.

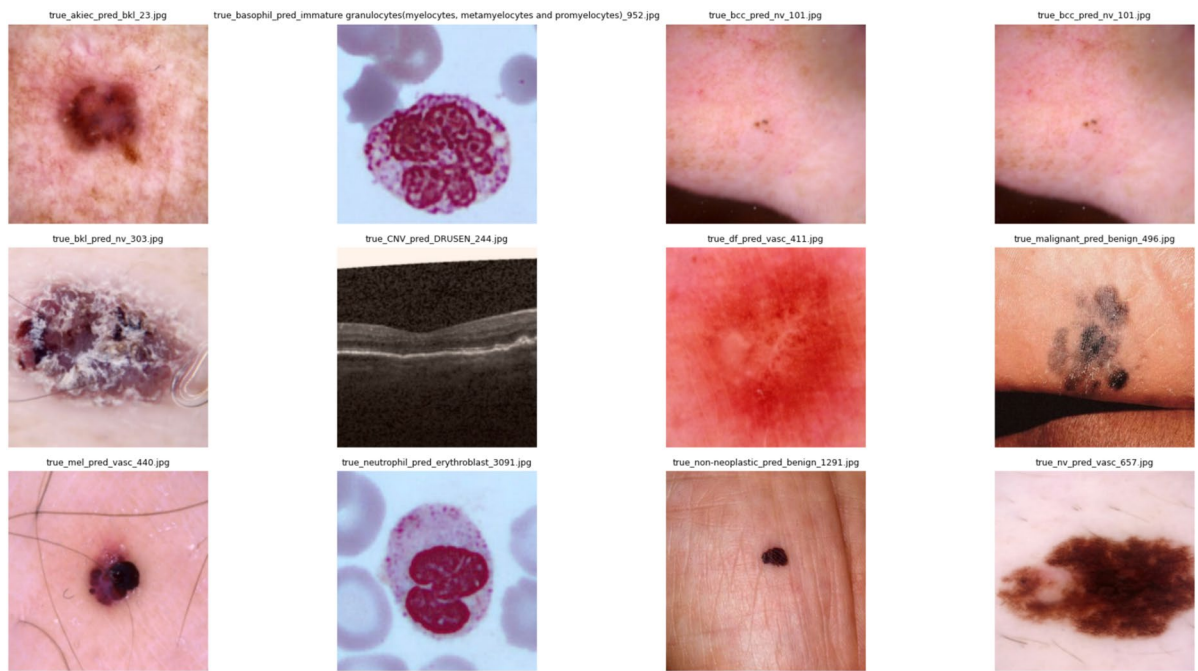


Fig. 13. Examples of where the model fails.

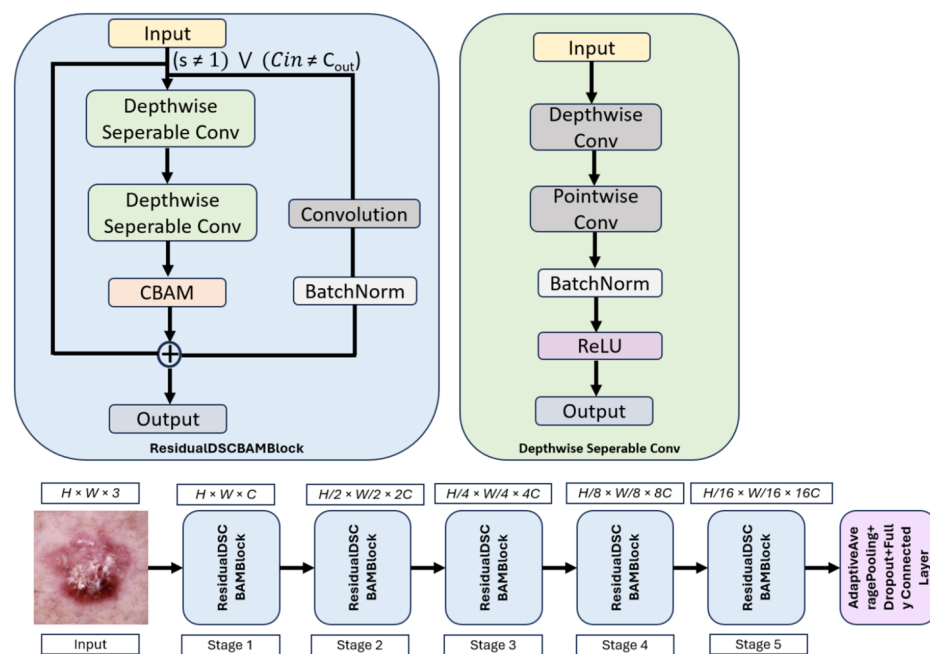


Fig. 14. Overview of the complete architecture. In this context, s denotes the stride of the convolution layer, while H , W , and C represent the height, width, and number of channels of the input image, respectively.

Stage	Filters	Stride	Layer / Block
Stage 1 (3 × 64)	64	1	ResidualDSCBAMBlock
Stage 2 (64 × 128)	128	2	ResidualDSCBAMBlock
Stage 3 (128 × 256)	256	2	ResidualDSCBAMBlock
Stage 4 (256 × 512)	512	2	ResidualDSCBAMBlock
Stage 5 (512 × 1024)	1024	2	ResidualDSCBAMBlock
Pooling	1024	–	AdaptiveAvgPool
Fc1	512	–	Fully Connected (Dense)
Fc2	n_c	–	Fully Connected (Dense)

Component	Type	ResidualDSCBAMBlock	Padding	Activation
		Kernel Size		
Conv1	Convolution	3 × 3	1	ReLU
Conv2	Convolution	1 × 1	0	ReLU
Bn	Batch Normalization	–	–	–
CBAM	Attention Module	–	–	–

Table 9. Architecture of the proposed MedNet model. The network is composed of five sequential convolutional stages, each built from a ResidualDSCBAMBlock that integrates depthwise separable convolutions with the Convolutional Block Attention Module (CBAM). The first table summarizes the number of filters, strides, and layer types across stages, while the second table details the internal structure of the ResidualDSCBAMBlock. The final layers include an adaptive average pooling operation followed by two fully connected (Dense) layers, where n_c denotes the number of output classes.

Data availability

The datasets generated and analyzed during the current study are available in the Zenodo repository, at <https://doi.org/10.5281/zenodo.10519652> and <https://medmnist.com/>.

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Author contributions

M.F. conceived and designed the study. M.F. and M.E. collected the data. M.F., S.M., and M.E. performed the data analysis. M.F. contributed to methodology. M.F. and S.M. contributed to the provision of resources. M.F. prepared the original draft of the manuscript. M.F. and S.M. reviewed and edited the manuscript. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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